Sirpreet Dhillon

Software Developer

Phone 236 869 5924

Email sirpreet.dhillon@gmail.com

SUMMARY

Software and AI Engineer with over 2 years of experience in building scalable data infrastructure and autonomous systems. Proven expertise in Python, C++, and Cloud Architecture (AWS), with a strong focus on optimizing data pipelines and deploying machine learning models into production environments.

PROFESSIONAL EXPERIENCE

Software Infrastructure Developer

True Solutions LLC

Sep 2024 – Dec 2025

- Architected and deployed data-driven web applications, designing custom CMS solutions that improved content management efficiency for internal teams.
- Engineered automated data pipelines to process large-scale datasets, ensuring seamless integration between backend databases and frontend user interfaces.
- Built internal workflow management tools using React and Node.js streamlining organizational processes and reducing manual data entry.

Data QA

Functional & Anatomical Imaging & Shape Analysis Lab

May 2021 – Dec 2021

- Designed and maintained robust ETL (Extract, Transform, Load) pipelines to manage and clean complex medical imaging datasets, achieving 100% data accuracy for research usage.
- Scripted automated reporting tools to visualize data quality metrics, significantly reducing the time required for manual validation cycles.
- Mentored junior team members on data cleaning protocols and best practices for dataset management.

Software Infrastructure Developer

Ford Motor Company

Sep 2019 – Dec 2019

- Developed automation scripts using Python and Bash to replace manual testing processes, achieving a 6.25% reduction in testing runtime across the development cycle.
- Performed root-cause analysis on complex software errors within an agile environment, deploying patches that ensured high availability for industrial engineering tools.

Research Assistant

Simon Fraser University

Sep 2017 – Dec 2018

- Engineered precise tracking systems and optimized control software for accurate data collection.
- Led research and development initiatives, designing efficient systems and automating testing protocols to improve performance.

PROJECTS

Deep Learning & Neural Network Optimization

Deep Learning | Neural Networks

Nov 2024

- Built and trained deep neural networks from scratch using python and NumPy, implementing forward and backward propagation algorithms to classify image data.
- Optimized model performance by implementing hyperparameter tuning strategies, including learning rate decay and regularization (L2/Dropout) to minimize overfitting.
- Structured a machine learning project pipeline, performing error analysis to identify data mismatches and improve model generalization on test sets.

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Autonomous Navigation System

Indoor Localization & Mapping | ROS | IoT | C++

Jan 2022 – Aug 2022

- Developed the complete autonomy software stack using ROS (Robot Operating System) and C++, enabling the robot to operate independently in dynamic environments.
- Implemented SLAM (Gmapping) algorithms to process real-time LiDAR data, achieving 90% localization accuracy and efficient path planning.
- Programmed low-level hardware interfaces and communication protocols to synchronize sensor data with motor control logic for precise navigation.

Real-Time Customer Retention Analytics

AWS | Data Analysis

Aug 2024

- Architected a serverless data pipeline using AWS Lambda and Kinesis, enabling real-time ingestion and processing of customer engagement metrics.
- Integrated AWS analytics tools to generate actionable insights on customer retention, visualizing data trends for business stakeholders.

5G V2I Network Simulation

Network Protocols | C++ | NS3

Jan 2021 – Apr 2021

- Modeled Vehicle-to-Infrastructure (V2I) communication systems using C++ and NS3 simulator, analyzing packet loss and latency under high-load conditions.
- Optimized network protocol parameters to improve system throughput in dynamic, high-mobility simulation environments.

SKILLS

Languages
- Python (Proficient), C++, SQL, Bash Scripting, XML

AI & Machine Learning
- TensorFlow, Keras, PyTorch, NumPy, Pandas, Hyperparameter Tuning

Cloud & DevOps
- AWS, Linux/Unix OS, CI/CD Pipelines, Git, Jenkins, JIRA

Robotics & Embedded Systems
- ROS (Robot Operating System), MQTT, I2C/SPI/UART Protocols

Microcontroller Experience
- Raspberry Pi, Arduino, Nvidia Jetson Nano, ESP32

Soft Skills
- Problem-solving, Attention to Detail, Communication, Time Management

EDUCATION

Bachelor of Applied Science | Computer Engineering
Simon Fraser University | Graduated: 2023

ADDITIONAL INFORMATION

Certifications
- **AWS Certified Cloud Practitioner** | Amazon Web Services | April 2024

Online Courses
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, Optimization** | DeepLearning.AI | Apr 2025
- **Structuring Machine Learning Projects** | DeepLearning.AI | Apr 2025
- **Neural Networks and Deep Learning** | DeepLearning.AI | Nov 2024